Multivariable Calculus (MT1008)

Date: 08th April, 2026

Course Instructor(s)

Dr. Akhlaq Ahmad Bhatti (Moderator)
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Sessional-I1

Total Time: 1 Hours

Total Marks: 65

Total Questions: 5

Section _____ Student Signature _____ Roll No _____

Important Instruction: 1. Only simple calculator is allowed. Use of programmable calculator or any helpful devices will result **F** grade in the course.

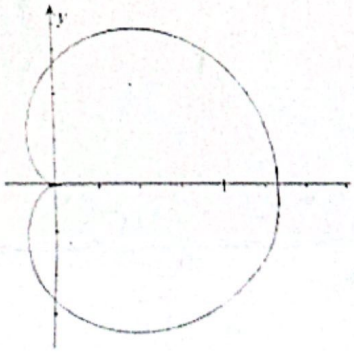
2. Question # 1(f) must be returned to the invigilator along with answer sheet.

CLO #2: Evaluation of Multiple Integrals in Different Coordinate Systems and Their Applications to Work, Circulation, Flux, Green's Theorem, and Stokes' Theorem.

Question # 1(a)[10]: Set up **(ONLY)** the below Cartesian integral into an equivalent Polar integral.

$$\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{2}{(1+x^2+y^2)^2}\, dy\, dx.$$

Question # 1(b)[10]: Find the area of the region **(sketch the region on answer sheet)** that lies inside the cardioid $r = 1 + cos\theta$ **(polar graph given below)** and outside the circle $r = 1$.



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Question # 1(c)[10]: Use the transformation $u = x - y, v = 2x + y$ to evaluate the integral

$$\iint_R (2x^2 - xy - y^2)dx\,dy,$$

for the region R in the first quadrant bounded by the lines $y = -2x + 4, y = -2x + 7, y = x - 2$, and $y = x + 1$.

Question # 1(d)[10]: Set up **(ONLY)** triple integral for the volume of the sphere $\rho = 2$ in **(a)** spherical, **(b)** cylindrical coordinates.

Question # 1(e)[10]: Find the line integral of $f(x, y) = ye^{x^2}$ along the curve

$$\boldsymbol{r}(t) = 4t\,\mathbf{i} - 3t\,\mathbf{j}, 1 \leq t \leq 2.$$

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